

Founding Family Heritage, Social Background and Risk Taking in Family Firms

11th Emerging Markets Finance Conference, 2020

Rama Seth

Department of Finance
Copenhagen Business School
rs.fi@cbs.dk

Ankit Singhal*

Doctoral Student
Shiv Nadar University
as559@snu.edu.in

Vishwanatha S R

Department of Finance
Shiv Nadar University
vishwanath.sr@snu.edu.in

December 15, 2020

*Presenting author

SHIV NADAR UNIVERSITY

Introduction

Do family firms take less or more risk?

- Does the involvement of family members influence corporate risk-taking?
- Presence of large shareholders influence risk-taking choices
- Financing perspective
 - family ownership is associated with lower cost of debt financing (Anderson, Mansi and Reeb,2003).
 - credit markets are likely to provide more long-term debt to family firms (Crocì et al.,2011).
 - religious founders take less risk than others (Jiang et al.,2015).
- Investment perspective
 - diversified large shareholders undertake riskier investments than firms controlled by non-diversified large shareholders (Faccio et al.,2011).
 - socioemotional reasons(Gómez-Mejía et al., 2007).
 - degree of CEO influence impacts variability in firm performance (Adams et al.,2005).
- Conflicting evidences
 - no relation (John, Litov, and Yeung,2008,Anderson and Reeb,2003).

- Blind application of ownership concentration on risk-taking is likely to yield inconsistent/ contrary results.
- All prior studies have been carried out incorporating market competition but their focus is not on family firms.
- Other studies indicates to a particular macroeconomic or personal event, or the manager's gender has an effect on risk-taking by the CEO.
- The way firms undertake risk is not only motivated by the ownership but external forces such as product market competition, cultural heritage and also biological disposition.

We are first to test the implications of CEO's social background on risk taking.

Background in India

- 73% of firms in the BSE 500 index are family run
- Unusually high average level of equity ownership by the family, 49%, as compared to about 18% in the U.S., 38% in Europe and 6% in Japan..
- less developed institutions and capital markets and poor legal protection for external investors (Khanna and Palepu, 2000),
- Founders have both the incentive to take risk as well as be risk averse.
- As manager-shareholder ownership increases, there is less incentive to extract private benefits (Jensen and Meckling, 1976)

What we are doing?

- Ownership, governance, management succession affects risk choices.
 - CEO's ability to influence corporate decisions is high.
- Ethnic background of founders.
 - individuals' attitude towards risk and uncertainty is at least partly shaped by cultural heritage (Chen, 2013; Becker, Dohmen, Enke, and Falk, 2015).
 - Indian family firms are loosely similar to LBO partnerships in the U.S.A
 - Families with a long history of being in business may be better positioned to manage risk
 - Firms belonging to the older business families to be more stable i.e. less risky, especially in terms of cash flow volatility.

What we are doing?- continued

- Biologically predisposed behaviors, culturally transmitted preferences are fundamental to understanding an individual's economic decisions (Giavazzi et al., 2019).
- Risk culture can be traced back to its beginning and to the people who founded the firm (Guiso et al. (2015a)).
- Executive's social class has a lasting impact on his/her tendency to take risk (Kish-Gephart and Campbell, 2015).
- Family-firms founded by religious founders take less risk than other family-firms (Hilary and Hui, 2009, Baxamusa and Jalal, 2016).
- Indian society is organised into castes(or communities) that segregates the society into four groups
 - 1 Brahmin (the priest class who officiate at religious ceremonies)
 - 2 Kshatriya (soldiers and kings)
 - 3 Vaishya (the business class)
 - 4 Sudra (those who undertake manual labor)
- Firms that are founded by individuals from business(ethnic) communities are more likely to take risk.

H1: We expect competition and growth prospects to play a moderating role.

Family Involvement

- Diversified shareholders have the incentive to expropriate bondholder wealth by investing in risky, high return projects (Jensen and Meckling, 1976)
- large undiversified ownership stakes have different incentives than small shareholders (Shleifer and Vishny, 1997).
- Preservation of socio-emotional wealth hypothesis

H2: It is likely that there is a curvilinear relation between family ownership and risk taking

Family Ownership

- Degree of ownership results in better alignment of interests and also provides insufficient incentives to undertake significant risk
- extent of resources diverted depends on the degree of investor protection in the country (Shleifer and Wolfenzon, 2002)
- Banks may limit risk taking by firms to protect their investment (John et al. 2008).
- individuals are more optimistic about outcomes which they believe are under their control (see Malmendier and Tate, 2005 and the papers cited therein)

H3: We hypothesize that firms owned by older families take less risk.

Family History

- preservation of inter-generational transfer hypothesis

H4: We hypothesize that firms headed by family members belonging to a business caste would be riskier.

Social Background

- Prior research shows that religion matters for corporate decision making (e.g. Hillary and Hui, 2009, Miller and Hoffman, 1995 and Miller, 2000, Jiang et al. 2015)
- Heterogeneity is high in India and identified by the surname

Preview of Main Findings from the study

- Family firms have lower equity, cash flow and earnings volatility after controlling for competition and growth opportunities, especially when family members are in leadership position.
- This is true of firms managed by older business families as well
- Some evidence that firms managed by family CEOs belonging to a business caste are riskier
- Endogeneity is a concern between family ownership and risk taking
 - Instrumental variable regression
 - Propensity score matching
 - Firm fixed effects
- Our results are generally robust to the use of alternate methodologies.
- We explain our results by examining the extent of diversification, leverage choices and cash holdings of family and non-family firms.

- Our initial sample consisted of all firms listed on the National Stock Exchange (NSE) during 2001-2015.
- Excluded - Banking, Insurance and Financial firms, Foreign firms
- Final sample consisting of 510 firms representing all industry groups such as manufacturing (75%), Information and communication (3%), construction and real estate (5%), wholesale and retail trading (8%), accommodation and food services (3%), agriculture (2%), transport (1%) and others (3%).
- Annual data on firm characteristics, ownership, governance and accounting performance from the CMIE Prowess database and stock market data from the National Stock Exchange of the India website.

- **Dependent variable** - Risk-Taking
 - S.D of monthly stock returns (Cain and McKeon, 2016)
 - S.D of cash from operations scaled by total assets
 - S.D of ROA (EBITDA/TA)*100 (Faccio et al. 2016; John, Litov and Yeung, 2008)
- Eleven 5-year, overlapping, windows (2001-2005, 2002-2006, 2003-2007, 2004-2008, 2005-2009, 2006-2010, 2007-2011, 2008-2012, 2009-2013, 2010-2014 and 2011-2015).
- **Explanatory variables**
 - Family dummy, Family CEO, Family CEO*Business caste, Family CEO*Old Business family

- **Control variables**

- $\ln(\text{TA})$, firm size, leverage, ROA, HHI, market-to-book ratio(M/B), Sales growth rate, Institutional shareholdings, Independent directors (%)

- **Instrumental variables (IV)**

- Family ownership on the natural log of total assets, the square of the natural log of total assets and Beta

- **Propensity Score Matching**

- Method: Nearest neighbor matching with replacement with caliper set equal to 0.2
- Treated group: Family dummy, Family CEO, Family CEO*Business caste, Family CEO*Old Business family
- Control group: Counterparts

Summary Statistics

Variables	Full sample				Family firms	Non-family firms	Difference t-statistic
	Mean	Std.Dev.	Min	Max			
ROA	0.12	0.05	0.03	0.23	0.12	0.11	3.56***
M/B	1.50	1.53	0.09	5.85	1.52	1.43	2.11**
Tangibility	0.35	0.18	0.05	0.67	0.35	0.34	1.54
HHI	791	1067	180	4150	794	780	0.45
Leverage	0.73	0.10	0.56	0.95	0.73	0.75	-6.10***
Institutional shareholding (%)	7.66	9.43	0.01	31.51	7.71	7.45	0.89
Independent directors (%)	0.27	0.16	0	0.45	0.28	0.25	5.81***
Diversification	0.23	0.42	0	1	0.24	0.20	3.48***
Tobin q	1.49	1.51	0	6.06	1.47	1.56	-2.10**
ln (assets)	7.56	1.63	4.78	10.65	7.58	7.50	1.56
ln(firm age)	3.59	0.39	3.04	4.41	3.60	3.55	4.32***
Sales growth rate	11.36	24.56	-39	65.42	11.53	10.73	1.10
$\sigma(\text{CFO})$	0.07	0.48	0.01	0.20	0.077	0.081	-0.86
$\sigma(\text{ROA}) \times 100$	0.006	0.009	0	0.03	0.006	0.007	-3.75***
$\sigma(\text{Stock returns})$	39.95	53.78	0.97	206.54	38.78	44.56	-3.61***
Family shareholding	42.34	24.90	0	74.85	53.40	0	NA

Volatility of Stock Returns

Family Dummy	-13.803 (-1.260)				
Family Shareholding		-1.443*** (-2.654)			
Sq. of family shareholding		0.023*** (3.108)			
Family CEO			-37.122*** (-4.074)		
Family CEO*Business caste				30.692*** (-3.092)	
Family CEO * Old Business family					-36.883*** (-3.425)
Ln (Firm age)	-20.844* (-1.919)	-22.635** (-2.085)	-17.586 (-1.619)	-13.284 (-1.238)	-15.464 (-1.409)
ln (Assets)	9.432*** (2.672)	8.614** (2.435)	9.527*** (2.706)	9.897*** (2.839)	9.559*** (2.712)
Leverage	-14.333 (-0.664)	-12.611 (-0.584)	-11.797 (-0.549)	-10.892 (-0.522)	-12.291 (-0.572)
ROA	27.172 (0.626)	25.156 (0.580)	26.626 (0.615)	-8.154 (-0.193)	27.941 (0.645)
Low HHI	18.158 (1.482)	20.650* (1.682)	18.868 (1.544)	11.842 (0.963)	17.465 (1.429)
High M/B	-9.125 (-0.971)	-10.832 (-1.151)	-8.182 (-0.872)	-12.737 (-1.367)	-8.315 (-0.886)
Sales growth rate	-0.023 (-0.297)	-0.026 (-0.332)	-0.018 (-0.231)	-0.016 (-0.205)	-0.018 (-0.236)
Institutional shareholding %	-0.492 (-0.956)	-0.065 (-0.121)	-0.363 (-0.706)	-0.280 (-0.551)	-0.403 (-0.785)
Independent directors (%)	-6.576 (-0.262)	-5.375 (-0.214)	-13.480 (-0.535)	-13.269 (-0.529)	-8.079 (-0.323)
Intercept	88.997* (1.819)	90.992* (1.857)	70.976 (1.445)	62.087 (1.289)	63.769 (1.307)
Observations	3697	3693	3697	3477	3697
Adj. R-squared	0.004	0.007	0.009	0.006	0.007
Inflection Point		30.78%			

Volatility of Cash Flows

Family Dummy	-0.012*** (-5.037)				
Family Shareholding		-0.001*** (-4.370)			
Founder CEO			-0.010*** (-5.340)		
Family CEO * Business caste				-0.004** (-2.249)	
Family CEO * Old Business					-0.017** (-2.178)
Ln (Firm age)	-0.006*** (-2.808)	-0.007*** (-2.952)	-0.007*** (-3.262)	-0.007*** (-2.946)	-0.007*** (-3.258)
Ln (Assets)	-0.005*** (-6.774)	-0.005*** (-6.862)	-0.005*** (-6.844)	-0.005*** (-6.370)	-0.005*** (-6.905)
Leverage	0.012** (2.497)	0.012** (2.555)	0.012*** (2.582)	0.014*** (2.968)	0.014*** (2.971)
ROA	0.010 (1.036)	0.010 (1.033)	0.009 (0.950)	0.010 (1.123)	0.010 (1.117)
Low HHI	-0.012*** (-4.385)	-0.011*** (-4.268)	-0.012*** (-4.477)	-0.013*** (-4.724)	-0.013*** (-4.751)
High M/B	0.007*** (3.541)	0.007*** (3.435)	0.007*** (3.756)	0.006*** (3.020)	0.007*** (3.469)
Sales growth rate	0.000* (1.936)	0.000* (1.915)	0.000* (1.916)	0.000* (1.814)	0.000** (1.968)
Inst. Shareholdings	-0.000* (-1.906)	-0.000 (-1.560)	-0.000* (-1.849)	-0.000* (-1.788)	-0.000 (-1.612)
Independent directors (%)	0.003 (0.528)	0.003 (0.547)	0.004 (0.789)	-0.001 (-0.211)	0.001 (0.278)
Intercept	0.140*** (13.566)	0.141*** (13.573)	0.143*** (13.755)	0.134*** (12.645)	0.135*** (13.120)
Observations	4111	4106	4111	3870	4111
Adj. R-squared	0.043	0.042	0.046	0.037	0.038

Volatility of Earnings

Family dummy	-0.011*** (-3.969)					
Family Shareholding		-0.000*** (-2.752)	-0.000*** (-3.445)			
Sq. Family Shareholding			0.000*** (2.684)			
Family CEO				-0.006** (-2.523)		
Founder*Business caste					-0.003 (-1.170)	
Founder* Old business						0.063*** (7.068)
Ln (firm age)	0.001 (0.188)	0.000 (0.089)	0.000 (0.076)	-0.001 (-0.267)	-0.000 (-0.079)	0.001 (0.265)
Ln (Assets)	-0.011*** (-13.575)	-0.011*** (-13.459)	-0.011*** (-13.612)	-0.011*** (-13.617)	-0.012*** (-13.271)	-0.011*** (-13.537)
Leverage	0.008*** (4.778)	0.008*** (4.826)	0.008*** (4.792)	0.008*** (4.873)	0.008*** (4.800)	0.008*** (4.980)
Low HHI	0.000 (0.001)	-0.000 (-0.129)	0.000 (0.081)	-0.000 (-0.142)	-0.001 (-0.376)	0.001 (0.387)
High M/B	-0.003 (-1.484)	-0.003 (-1.413)	-0.004 (-1.563)	-0.003 (-1.477)	-0.004 (-1.605)	-0.004* (-1.684)
Sales growth rate	-0.000 (-0.005)	0.000 (0.011)	-0.000 (-0.021)	-0.000 (-0.020)	0.000 (0.044)	0.000 (0.001)
Inst. Shareholding	0.001*** (7.699)	0.001*** (7.242)	0.001*** (7.645)	0.001*** (7.683)	0.001*** (7.596)	0.001*** (7.537)
Independent directors %	-0.003 (-0.545)	-0.004 (-0.660)	-0.003 (-0.536)	-0.003 (-0.501)	-0.005 (-0.711)	-0.006 (-1.027)
Intercept	0.094*** (8.334)	0.092*** (8.149)	0.095*** (8.360)	0.094*** (8.237)	0.094*** (7.864)	0.084*** (7.445)
Observations	4198	4193	4193	4198	3956	4198
Adj. R-squared	0.055	0.053	0.055	0.053	0.052	0.063
Inflection point			47%			

Instrumental Variables

Predicted value of Family Shareholding	-31.883** (-2.131)
Sq. of Family Shareholding	0.414** (2.157)
Ln (Firm age)	-12.203 (-0.774)
leverage	-82.083* (-1.778)
ROA	36.327 (0.610)
Low HHI	79.082** (2.451)
High M/B	-23.345 (-1.546)
Sales growth rate	0.002 (0.020)
Institutional Shareholdings	4.017** (2.285)
Independent directors (%)	81.321 (1.520)
Intercept	418.640*** (2.788)
Observations	3693
Wald Chi-Square	10.36
Inflection Point	38.49%

IV NABAR UNIVERSITY

Propensity Score Matching Results

Panel A: Stock Return Volatility

	N	Mean %	Difference %	t-statistic
σ (Stock returns) Family Firms	3249	71.67	-12.20	-0.77
σ (Stock returns) Non Family Firms		83.87		
σ (Stock returns) Family-CEO *Old Business Family	182	76.03	-14.26	-0.42
σ (Stock returns) control sample		90.30		
σ (Stock returns) Family-CEO*Business caste	2,039	68.63	7.20	0.68
σ (Stock returns) Control sample		61.44		

Panel B: Cash Flow Volatility

	N	Mean %	Difference %	t-statistic difference
σ (CFO) Family Firms	3,261	0.069	-0.0122	-3.29***
σ (CFO) Non Family Firms			0.081	
σ (CFO) Family-CEO	3,065	0.069	-0.0162	-4.78***
σ (CFO) Non family-CEO			0.085	

Propensity Score Matching Results

Panel C: ROA Volatility

	N	Mean %	Difference %	t-statistic difference
$\sigma(\text{ROA}) \times 100$ Family Firms	3,328	0.008	-0.009	-1.40
$\sigma(\text{ROA}) \times 100$ Non Family Firms			0.017	
$\sigma(\text{ROA}) \times 100$ Family-CEO	3,128	0.008	-0.005	-0.84
$\sigma(\text{ROA}) \times 100$ Non family-CEO			0.013	
$\sigma(\text{ROA}) \times 100$ Family-CEO * Old Business Family	1,476	0.017	0.001	-0.27
$\sigma(\text{ROA}) \times 100$ Family-CEO * Control Sample			0.012	

Volatility of Stock Returns - Firm Fixed Effects

Family Dummy	-16.999 (-0.705)			
Family CEO		-34.228* (-1.730)		
Founder*Business caste			71.456*** (3.495)	
Family CEO*Old Business				-41.975* (-1.705)
Independent directors (%)	-3.004 (-0.055)	-7.592 (-0.139)	2.845 (0.045)	-4.966 (-0.091)
Ln (Firm age)	-18.090 (-0.752)	-15.624 (-0.649)	-16.916 (-0.624)	-13.889 (-0.569)
Ln (Assets)	-1.670 (-0.253)	-1.670 (-0.253)	-6.342 (-0.962)	0.092 (0.014)
Leverage	7.859 (0.431)	7.859 (0.431)	7.681 (0.434)	7.932 (0.435)
ROA	-3.598 (-0.108)	-3.598 (-0.108)	2.943 (0.091)	-1.953 (-0.059)
Low HHI	-22.322 (-0.815)	-22.322 (-0.815)	-24.714 (-0.908)	-23.355 (-0.852)
High M/B	1.529 (0.201)	1.529 (0.201)	0.175 (0.023)	1.673 (0.220)
Sales Growth rate	0.015 (0.269)	0.015 (0.269)	0.011 (0.197)	0.014 (0.253)
Institutional Shareholding	0.206 (0.367)	0.206 (0.367)	0.211 (0.380)	0.210 (0.374)
Intercept	75.352 (0.641)	59.312 (0.503)	59.324 (0.449)	49.360 (0.422)
Observations	3697	3697	3477	3697
Log-likelihood	2.493e+04	2.493e+04	2.339e+04	2.493e+04
Chi-square	6.6071	9.3414	20.0634	10.4937

Volatility of Cash Flows - Firm Fixed Effects

Family dummy	-0.010*			
	(-1.729)			
Family CEO		-0.009**		
		(-2.002)		
Family CEO *Business caste			-0.002	
			(-0.348)	
Founder*Old business family				-0.027
				(-1.421)
Independent directors (%)	0.004	0.005	-0.001	0.003
	(0.271)	(0.375)	(-0.064)	(0.199)
ln(firm age)	-0.007	-0.008	-0.007	-0.008
	(-1.228)	(-1.373)	(-1.145)	(-1.407)
ln(Assets)	-0.004***	-0.004***	-0.004***	-0.004***
	(-3.023)	(-3.023)	(-3.153)	(-3.023)
Leverage	-0.012***	-0.012***	-0.011***	-0.012***
	(-3.011)	(-3.011)	(-2.898)	(-3.011)
ROA	-0.020***	-0.020***	-0.020***	-0.020***
	(-2.897)	(-2.898)	(-2.940)	(-2.897)
Low HHI	-0.021***	-0.021***	-0.014***	-0.021***
	(-4.031)	(-4.031)	(-2.712)	(-4.031)
High M/B	0.006***	0.006***	0.006***	0.006***
	(4.163)	(4.163)	(4.022)	(4.163)
Sales growth rate	0.000***	0.000***	0.000***	0.000***
	(4.433)	(4.433)	(4.370)	(4.433)
Inst. Shareholdings	-0.000***	-0.000***	-0.000***	-0.000***
	(-3.383)	(-3.383)	(-3.310)	(-3.383)
Intercept	0.099***	0.102***	0.089***	0.093***
	(3.570)	(3.685)	(3.122)	(3.381)
Observations	4111	4111	3870	4111
Log-likelihood	6928.4400	6930.1348	6534.2418	6927.9578
Chi-square	131.1140	134.7896	116.7237	130.0646

Volatility of Earnings - Firm Fixed Effects

Family dummy	-0.024*** (-5.020)				
Family Shareholding		0.001*** (2.667)			
Family CEO			-0.010*** (-2.952)		
Family CEO.*Business caste				-0.002 (-0.249)	
Family CEO *Old business					0.055*** (4.405)
Independent directors %	0.005 (0.532)	0.003 (0.274)	0.001 (0.064)	0.002 (0.187)	-0.002 (-0.223)
Ln (firm age)	0.004 (1.030)	0.004 (0.878)	0.003 (0.783)	0.003 (0.656)	0.003 (0.678)
Ln (Assets)	-0.010*** (-5.025)	-0.011*** (-5.094)	-0.010*** (-4.981)	-0.011*** (-4.912)	-0.010*** (-4.950)
Leverage	0.005*** (2.762)	0.005*** (2.741)	0.005*** (2.738)	0.005*** (2.658)	0.005*** (2.721)
Low HHI	-0.002 (-0.240)	-0.002 (-0.285)	-0.002 (-0.238)	-0.001 (-0.154)	-0.002 (-0.236)
High M/B	-0.002 (-0.716)	-0.002 (-0.822)	-0.002 (-0.709)	-0.002 (-0.734)	-0.002 (-0.705)
Sales growth rate	0.000 (0.061)	0.000 (0.103)	0.000 (0.060)	0.000 (0.059)	0.000 (0.060)
Institutional Shareholdings	0.001*** (2.931)	0.001*** (3.465)	0.001*** (2.905)	0.001*** (2.899)	0.001*** (2.887)
Intercept	0.105*** (5.571)	0.098*** (5.430)	0.092*** (5.181)	0.098*** (5.198)	0.086*** (5.100)
Observations	4198	4193	4198	3956	4198
Log-likelihood	5279.7379	5266.6277	5268.6528	4849.2335	5273.2192
Chi-square	174.9574	193.7575	188.5071	179.6854	218.8785

Family Ownership, Corporate Diversification and Leverage

	ln (Number of business segments)	Leverage	
Family Shareholding	-0.002* (-1.786)	-0.001*** (-3.792)	
Sq. of Family Shareholding	0.000** (2.361)	0.000** (2.156)	
Founder CEO			-0.026*** (-4.033)
Ln (Firm age)	0.095*** (4.836)	0.006 (0.722)	0.002 (0.240)
Ln (Total Assets)	0.062*** (8.835)	0.011*** (4.410)	0.010*** (4.288)
Leverage	-0.086 (-1.030)		
Diversification dummy		-0.004 (-0.565)	-0.004 (-0.542)
ROA	-0.250** (-2.473)	-0.283*** (-9.244)	-0.288*** (-9.418)
Low HHI	-0.030 (-1.343)	0.021** (2.374)	0.020** (2.211)
High M/B	0.014 (0.791)	0.005 (0.834)	0.006 (0.850)
Sales growth rate	-0.000 (-0.591)	-0.000 (-0.236)	-0.000 (-0.250)
Inst. Shareholding %	0.000 (0.201)	-0.001*** (-3.075)	-0.001*** (-2.863)
Independent directors (%)	0.080 (1.635)	0.010 (0.568)	0.011 (0.641)
Intercept	0.146 (1.385)	0.695*** (21.054)	0.694*** (20.939)
Observations	1585	4112	4117
Adj. R-squared	0.104	0.036	0.033
Inflection point	27.04%	64.44%	

Nonlinear Relation Between Cash Flow Volatility and Family ownership

Family Shareholding	-0.001*** (-4.37)
Sq. of Family Shareholding	0.000*** (3.40)
ln (firm age)	-0.007*** (-2.952)
ln (Total Assets)	-0.005*** (-6.862)
Leverage	0.012** (2.555)
ROA	0.01 (1.033)
Low HHI	-0.011*** (-4.268)
High M/B	0.007*** (3.435)
Sales Growth Rate	0.000* (1.915)
Institutional shareholding	0.000 (1.560)
Independent directors (%)	0.003 (0.547)
Intercept	0.141*** (13.573)
Observations	4106
Adjusted R-Squared	0.042
Inflection Point	47.07%

Determinants of Cash Holdings

Family Shareholding	0.006*** (3.069)		
Sq (Family Shareholdings)	-0.000*** (-3.053)		
Family dummy		0.059 (1.471)	
Old business family			0.066* (1.939)
M/B ratio	0.001 (0.649)	0.001 (0.588)	0.001 (0.590)
Ln (Assets)	-0.097*** (-9.085)	-0.096*** (-8.982)	-0.097*** (-9.115)
R&D/Sales	-0.567 (-0.651)	-0.664 (-0.763)	-0.571 (-0.657)
Cash flow/Assets	0.362*** (5.184)	0.363*** (5.197)	0.361*** (5.174)
NWC/Assets	-0.276*** (-5.324)	-0.270*** (-5.217)	-0.267*** (-5.167)
Capex/Assets	-0.206** (-2.421)	-0.208** (-2.449)	-0.203** (-2.390)
Leverage/Total Assets	2.665*** (3.452)	2.697*** (3.495)	2.647*** (3.429)
Industry sigma	0.002*** (4.744)	0.002*** (4.683)	0.002*** (4.654)
Dividend dummy	0.500*** (13.458)	0.489*** (13.252)	0.491*** (13.339)
Intercept	-4.391*** (-21.481)	-4.385*** (-21.423)	-4.346*** (-21.506)
Observations	6835	6858	6858
Adj. R-squared	0.107	0.105	0.105
Time Fixed Effects	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes
Inflection point	56%		

CONCLUSION

- Relation between family ownership and risk is nonlinear.
- Firms run by family CEOs belonging to a business caste are riskier.
- Family firms account for much of entrepreneurial activity it is necessary that they do not take excessive risk
- Our study indicates that family firms are usually prudent in risk taking and that the social background of the founding family may be important in explaining volatility at the firm level.

The End